

HYPOTHESIS TESTING

CONTROL GROUP $\longleftrightarrow$ TREATMENT GROUP

BEFORE MANIPULATION $\longleftrightarrow$ AFTER MANIPULATION

QUESTION: HAS SOMETHING CHANGED?

H_0 : NULL Hypothesis (no change)

H_A : ALTERNATIVE Hypothesis

$\updownarrow$
NULL Hypo. is FALSE (change observed)

Actual Truth

	H_0 is TRUE	H_0 is FALSE
Reject H_0	TYPE I ERROR Probability p 5%	CORRECT
FAIL TO REJECT H_0	CORRECT	TYPE II ERROR Probability β

$p \rightarrow$ significance level

$1 - \beta \rightarrow$ "power" of the test

PROCEDURE

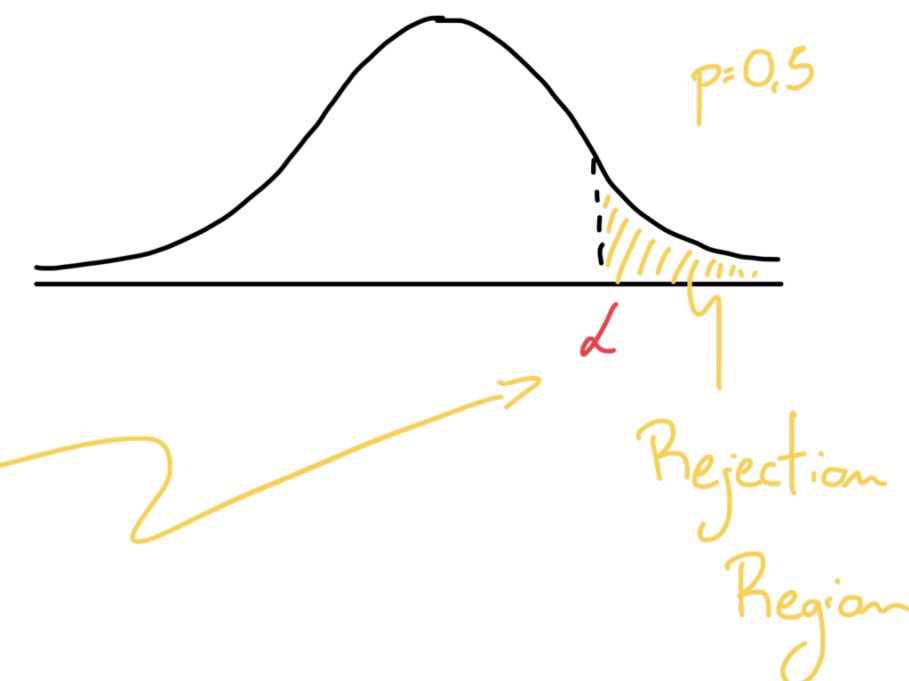
① We assume Null Hypothesis H_0 is true

② We define a test statistic Z

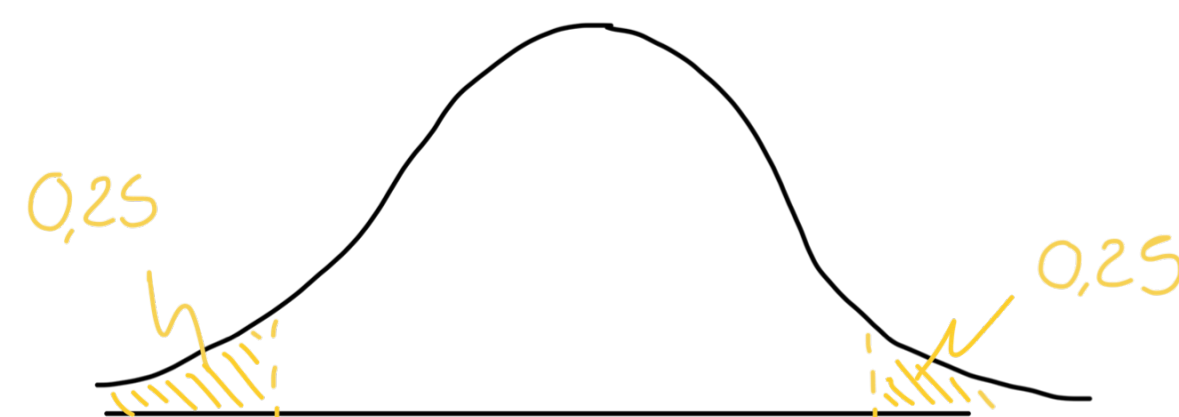
$$Z = \frac{\text{Obs } \bar{X} - \text{expected avg.}}{\text{Standard error}} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

③ Compute the observed significance level p .

How small of Z would reject the null hypothesis p is $P(Z \geq z)$ the probability that my Null Hypo. is True.



Choose $p=0.5$ rejection criteria



TWO SIDED REJECTION REGION

Decision